

Maximize the Topmost Element After K Moves

LeetCode 2202

American Express

Rules & Operations

We are allowed to perform only two types of operations on the pile:

- **Remove** the topmost element of the pile.
- **Add** any one element from the removed pit back to the pile to make it the topmost element.

Example Walkthrough

Input: Array = [2, 4, 7, 1, 8], k = 2

Steps:

- Remove 2 (pile = [4, 7, 1, 8], pit = [2])
- Remove 4 (pile = [7, 1, 8], pit = [2, 4])

Topmost becomes 7.

Max value possible = $\max(7, 2, 4) \rightarrow 7$ (Output)

Key Observations

- **Option 1:** After exactly k moves $\rightarrow$ remove exactly k elements. The top becomes `nums[k]` (if it exists).

- **Option 2:** Remove $k - 1$ elements and put back the largest removed element.

Edge Cases

Condition	Result
Only one element AND odd moves	-1
Only one element AND even moves	nums[0]
No moves ($k = 0$)	nums[0]
One move ($k = 1$)	nums[1]

C++ Solution

```
class Solution {
public:
    int maximumTop(vector<int>& nums, int k) {
        int n = nums.size();

        if(n == 1 && k % 2 == 1) {
            return -1;
        }

        if(n == 1 && k % 2 == 0) {
            return nums[0];
        }

        if(k == 0) {
            return nums[0];
        }

        if(k == 1) {
            return nums[1];
        }

        int ans = 0;
        for(int i = 0; i < min(n, k - 1); i++) { // pehle k-1 elements remove krke dekh
            ans = max(ans, nums[i]);
        }

        if(k < n)
            ans = max(ans, nums[k]); // kya max removed element abhi ke topmost se bada

        return ans;
    }
};
```